LLM-BASED GENERATION OF REALISTIC DOMAIN MODEL INSTANCES

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01 CONTEXT & MOTIVATION

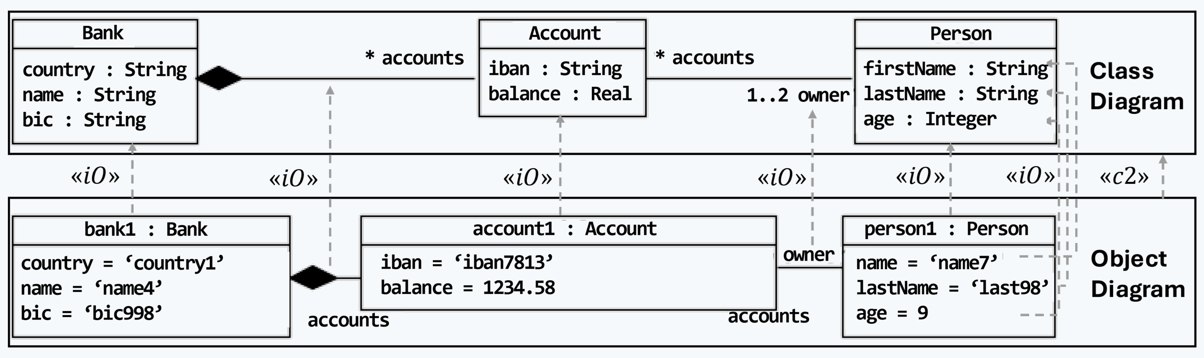
CHALLENGE

Validating software architectures requires analyzing specific system states using model instances.

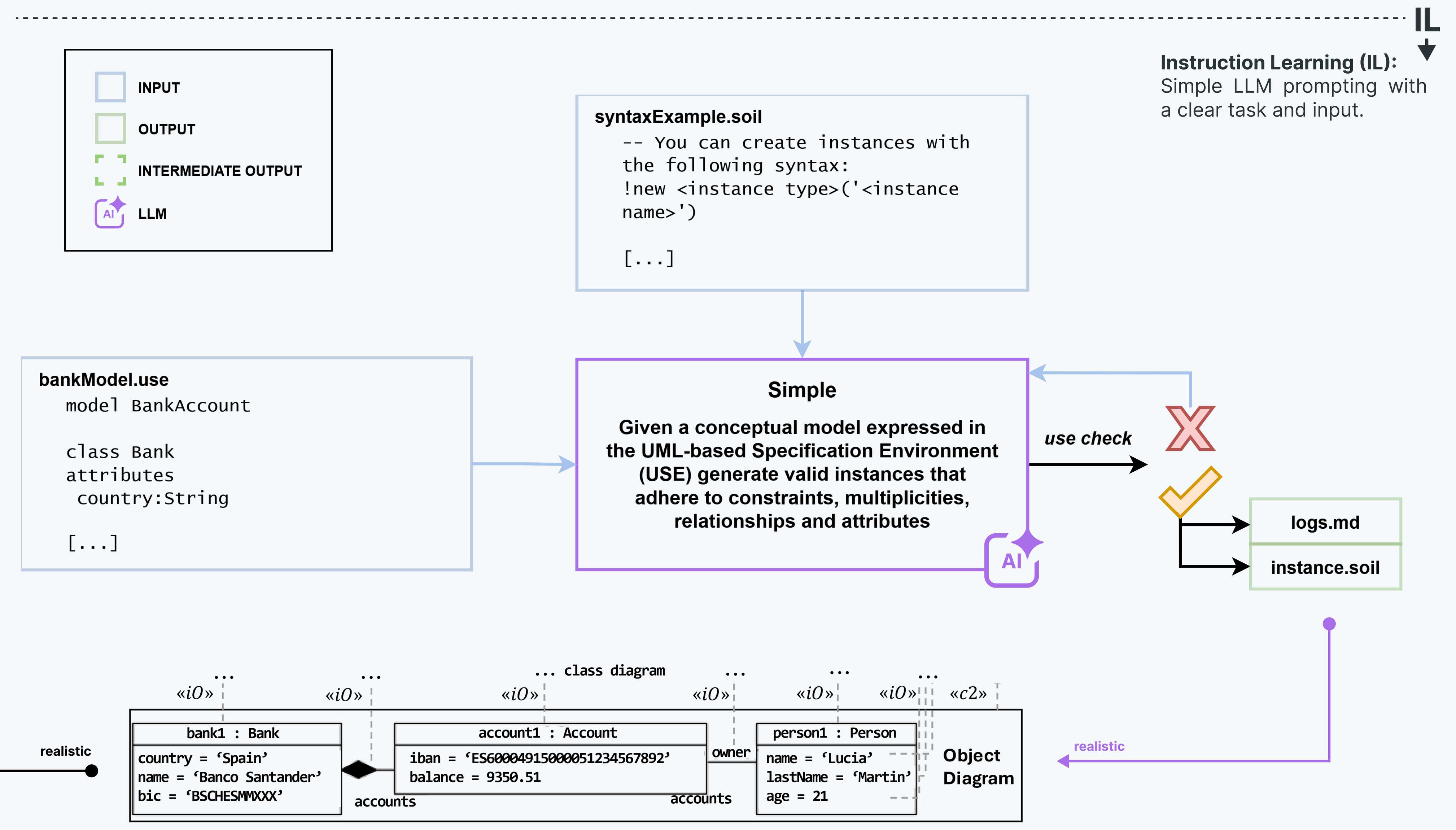
- **Manual generation** of these test inputs **is a tedious, time-consuming, and error-prone** process that often **suffers from human bias**.
- While **automated approaches** (e.g., formal methods or search-based techniques) **exist, they typically produce synthetic models** that lack real-world semantics.
- Furthermore, **current AI-assisted modeling research** primarily **focuses on completing type-level descriptions** (class diagrams), leaving the instance-level largely unexplored.

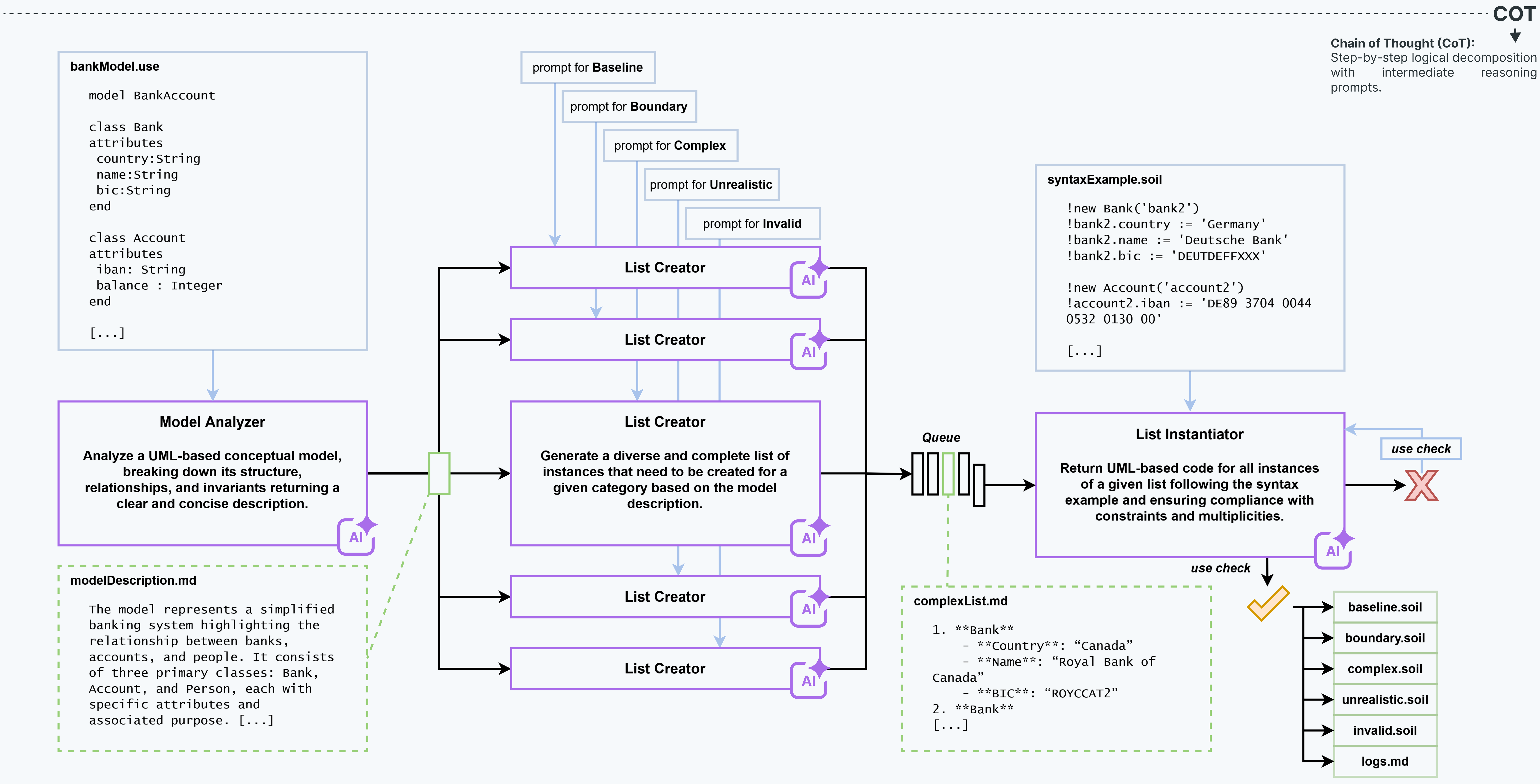
GOAL

Leverage Large Language Models (LLMs) to automatically **generate realistic instances from given domain models.**



02 METHODOLOGY





03 RESULTS

VALIDATION

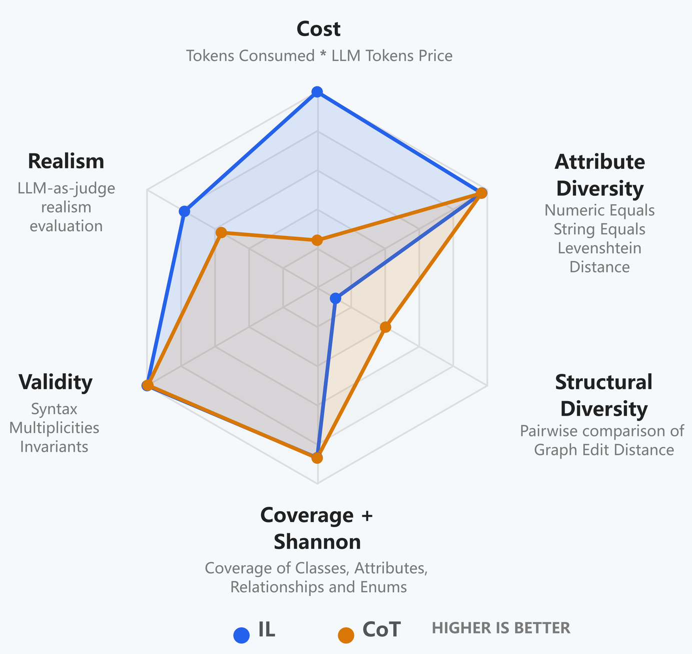
Both strategies proved exceptional at generating valid .soil instances. Syntax, Multiplicities, and Invariant conformance all scored **above 99.2%** across the board.

QUALITY & REALISM

The standard Instruction Learning (IL) approach achieved significantly higher real-world logic **(78.0%)** compared to Chain-of-Thought (56.3%), evaluated via a **Gemini 3.1 Pro LLM-as-a-Judge**.

MODEL COVERAGE & INSTANTIATION

The LLM successfully explores the structural blueprint. Both approaches utilized over **86% of available classes** and **>83% of attributes**, filling over 9,000 individual attribute slots with data.



04 CONCLUSION

We successfully demonstrate that LLMs can automate the generation of domain instances that are not only **syntactically flawless** but also **highly realistic.** We validated two distinct prompting strategies: **Instruction Learning (IL)** which provides a simple, efficient approach for **producing high-quality, realistic instances** and **Chain-of-Thought (CoT)**, which leverages multi-stage reasoning to explore diverse test scenarios, **proving superior for model validation** by producing less realistic, but larger and more structurally diverse instances that effectively identify **potential model under-specifications**.

NEXT STEPS

Our future work will focus on **scaling to enterprise models,** creating optimized **agent-based tooling,** and generate massive model corpora **to fine-tune specialized LLMs.**

A. Coman, L. Burgueño, D. Bork, and M. Wimmer, "Llm-based generation of semantically diverse and realistic domain model instances", Apr. 2026, arXiv:2604:10350 [cs.SE]. [Online]. Available: https://arxiv.org/abs/2604.10350